Bittle X Platform v2.0 - Interface Specification

Version 2.0, 2026-10-15. The platform is the stock Petoi Bittle X back cover with 4 teardrop pegs on its top. An add-on has a plate with 4 keyholes. Put it on the pegs, slide it 7 mm rearward, and it clicks. The keyholes have the shape of the pegs, so the add-on goes on in one direction only. No tools, glue or tape are necessary. interface.json has all the numbers in machine-readable form.

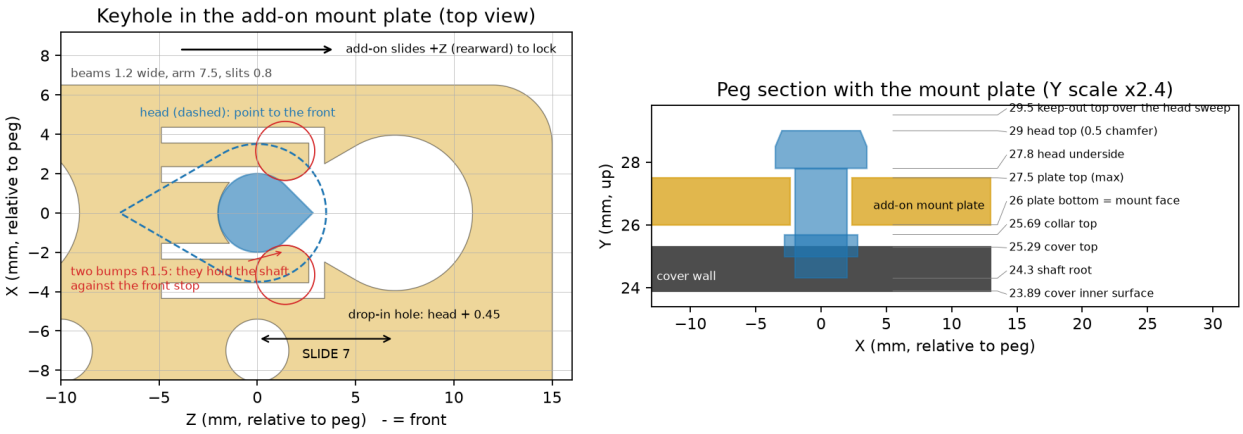
Version 2.0 is not compatible with version 1.x: the pegs have a new shape and a new place. Print the cover and the add-on plates again.



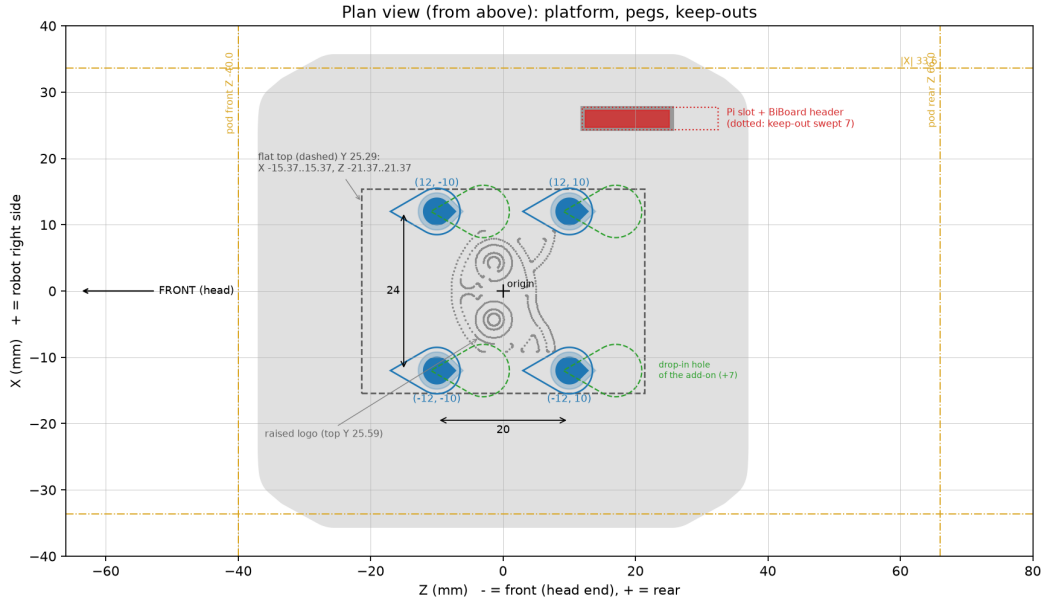
The platform (dark) with the starter plate (yellow).

1. Key numbers

Item	Value (mm)
Frame	X lateral (+X = robot right), Y up, Z along the body (-Z = head end). Origin: center line, middle of the cover.
Pegs (X, Z)	(12, -10), (12, 10), (-12, -10), (-12, 10): a centered 24 x 20 rectangle
Peg	teardrops. Head: R3.5, point 60 deg to the front (-Z), Y 27.8 to 29. Shaft: R2, point 90 deg to the rear (+Z)
Add-on plate	bottom at Y 26, 1.5 thick at the keyholes
Keyhole	slot 4.7 wide from the peg to 7 mm behind it (+Z), round front stop; drop-in hole at the rear end = the head teardrop + 0.45
Fit	put the add-on on 7 mm forward of its place, then slide it 7 mm rearward until it clicks
Click (detent)	two 1.2 wide spring beams, one on each side of each slot, freed by 0.8 U slits (fixed at the front end only). Bumps R1.5 at 1.42 behind the peg, 0.35 into the shaft path. They push the shaft against the front stop and center it.
Click force	about 4.2 N to slide on and 4.2 N to slide off (all 8 beams, PLA estimate); beam strain 1.12 %
Keep clear	the peg heads as they slide; the BiBoard header in the cover slot (X 24.73..27.27, Z 12.4..25.1, top Y 29.1)
Size	the pod fits in Z -40..66, |X| <= 33.6; outside that, check the legs and the head



Left: keyhole and click beams in the add-on plate (top view, locked). Right: the peg and the plate (side section).



Plan view: pegs (blue), drop-in holes (green), BiBoard header (red), pod size (gold).

2. Design an add-on

1 Start from cad/starter_plate.step. It has the keyholes and the click beams.

2 Add your part on top of the plate, in the same print. You can also screw parts to the M3 holes.

3 Keep your part out of stl/installed/keepout.stl. This keeps it clear of the robot, the peg heads and the header.

4 Keep heavy parts low and near the pegs. Keep the center of mass of the add-on within 10 mm of Z 0, near X 0.

5 Keep the weight low and near the center line. The pegs carry the whole add-on.

6 For a plate of a different shape: subtract cad/keyhole_cutter.step, then add cad/detent_bumps.step. Both are in position.

3. Printing

Part	Orientation	Supports	Cura profile
cover_pegs_foot (recommended)	stand on the rear end, foot on the bed	none: the foot is the support	Bittle Cover - PLA 0.25, brim off. Snap off the foot after the print.
cover_pegs (no foot)	stand on the rear end	tree, from the bed	Bittle Cover - PLA 0.25 (14-line brim)
starter_plate and add-ons	plate bottom on the bed	none	Bittle Pi Pod - PLA 0.25
small, detailed parts	-	-	Bittle Fine - PLA 0.10

Setting	0.25 profiles	Fine 0.10
Layer	0.25	0.10 (first layer 0.25)
Walls / infill	3 / 20 % gyroid	3 / 20 % gyroid
Outer wall / inner wall / infill speed	30 / 40 / 60 mm/s	25 / 35 / 50 mm/s
Outer wall acceleration	600 mm/s²	500 mm/s²
Nozzle	220 C (PLA)	220 C (PLA)

1. Material: PLA works. Use PETG for add-ons that you remove often or that get warm.
2. Standing on its end, the cover prints the pegs with layers along their length. This makes them strong. The shaft and collar points are down, so they need no supports.
3. The cover stands on a thin strip, and its center of mass is past that strip. The foot is a 3-layer plate with 7 fins under the leaning face. It increases the first layer from 53 to about 1000 mm². The fins touch the cover only at small teeth.
4. In add-ons, keep overhangs at 45 deg or less. Use chamfers, not round edges, on the bottom side.
5. The fits use 0.35 (slot), 0.45 (drop-in hole) and 0.3 (under the heads) clearance. Calibrate your printer if the plate does not slide on.

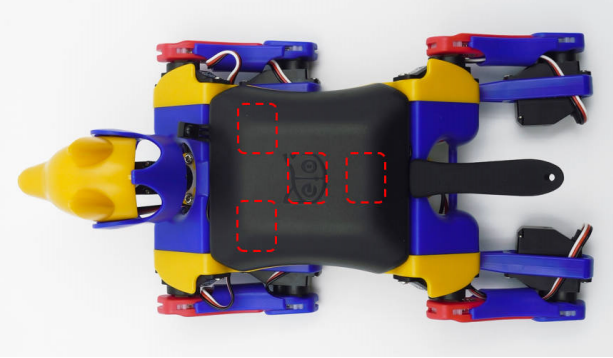
4. Files

File	Content
stl/print/	the parts to print, in print orientation
cad/starter_plate.step	the start point for an add-on
cad/pegs.step, keyhole_cutter.step, detent_bumps.step	the interface parts, in position (robot frame)
stl/installed/	all parts and keepout.stl in the robot frame
interface.json / interface.py	all numbers
build.py, docs.py	rebuild the parts and this document (build123d, manifold3d)

Open items: the BiBoard header height is an estimate from photos. The legs, the head and the robot balance are not modeled. The Petoi cover is a mesh, so cover_pegs is STL only.

5. Bittle X V2 back touch sensor

1. On a Bittle X V2, the back touch sensor is a flexible circuit that is glued to the inside of the cover top. It has touch areas on the left, the right, the center and near the tail (docs.petoi.com/extensible-modules/back-touch-sensor).
2. The peg roots stop inside the 1.4 mm wall, so the inside of the cover does not change. Move the circuit from the stock cover to cover_pegs in the same way as on the stock cover.
3. An add-on on the platform covers most of the touch areas. Disable the back touch mode (serial command Xb) when an add-on is on the robot.
4. For this reason, the platform has no parts under the cover top. A cover that prints flat without supports needs pegs held from inside the cover, where the circuit is. A possible interface v2 with sockets in a thicker top avoids this.



Bittle X V2 touch areas (red boxes). Image: Petoi, docs.petoi.com/extensible-modules/back-touch-sensor.

6. License

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